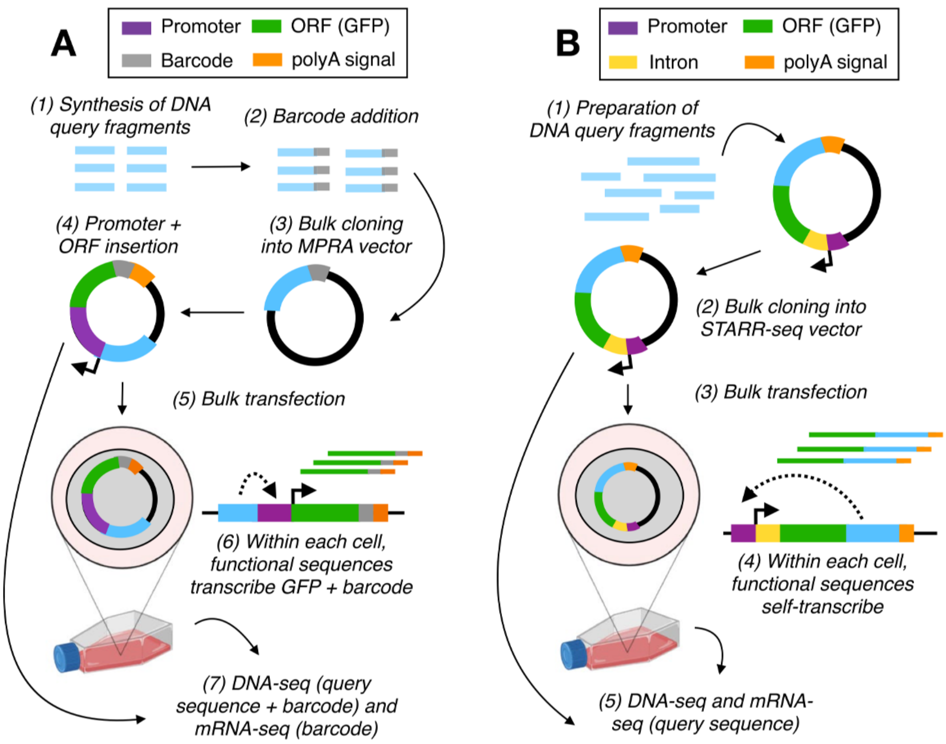
Massively parallel reporter assays: applications and QC

MPRAs are a complex family suite of assays that can be used to, broadly, interrogate the function of non-coding regions of the genome. Given the breadth of functions and roles the non-coding genome has, there are many different flavours of MPRAs and a comprehensive discussion of them all is outside the scope of this manual; there is a brief overview of some of these techniques in Gallego Romero & Lea, Genome Biology, 2023. Similarly, this breadth makes it hard to provide specific guidance that is suitable for all possible MPRA designs, and therefore all guidance here should be taken with a grain of salt, and fine-tuned to your specific circumastances. If in doubt, my advice is to err on the side of being statistically conservative.

All MPRAs began as an extension of the traditional plasmid luciferase or GFP reporter assay, where a single element of interest is inserted upstream of a minimal promoter and a fluorescent reporter, and activity is quantified by fluorescence levels. An MPRA appends a barcode to this reporter, and that is what is used to quantify abundance and regulatory potential of the tested element:



(Gallego Romero & Lea, Genome Biology, 2023)

Generic steps in an MPRA

A generic MPRA workflow involves the following steps. Every step, whether experimental or computational, will present its own challenges (and frustrations), and require extensive QC before you should feel comfortable moving forward to the next one.

1. Library design and assembly

2. Associating barcdoes to oligos (pre-transfection!)
3. Performing the actual MPRA
4. Raw data quantification (post-transfection!)
5. Barcode-level QC
6. Oligo-level QC
7. Identify active oligos/regions
8. Identify differentially active oligos/regions (contingent on study design)

The next sections discuss each of the above steps in variable levels of depth, providing additional context and external references when relevant. Given the breadth of designs, questions and approaches, there is currently no authoritative "Best practices" for MPRA design, but if you want to view a complete data analysis pipeline, the one in use by the upcoming IGVF release is available here. This pipeline combines a few tools that are discussed below.

Library design and assembly:

Will not be covered today. There are good primers for this elsewhere, eg Gordon et al, Nature Protocols, 2020, but the best way is to read papers that sort of do what you want to do and then extend or adapt their approach to your own question. Also, email people. The MPRA community is small and friendly, and generally appreciative of others trying to make these technologies work.

Associating barcodes to oligos:

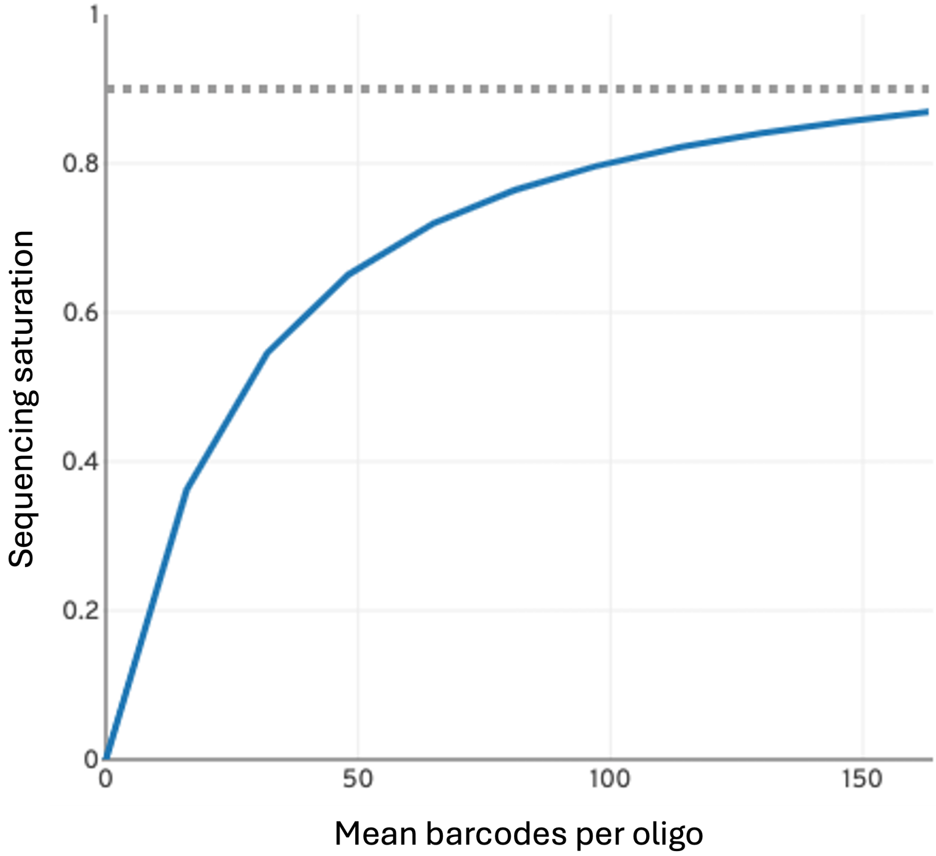
Will not be covered in-depth today, but this step is essential for assessing initial library complexity and ensuring good representation of original oligo pool. A rough rule is to aim for 80-200 unique barcodes per oligo *before* transfection so that you retain enough representation at the final stage (there's some variation here if your plasmid is), and to ensure that each barcode uniquely tags one element in your library. You can roughly control the number of barcodes in your input library by playing with the stoichiometry and the volume of bacteria in which you clone your plasmids.

This step should be performed *before* library transfection (and ideally before ORF and promoter insertion), because there's no point in transfecting a bad quality library. When carrying out this step, your assay type will determine the kind of sequencing set-up that you need—if you are testing different sequence variants within a small region you will likely need to sequence across the entire oligo using a longer paired-end design, but if you are testing very different regions you may be able to sequence only the barcode and the tail-end of your oligo using a shorter single-end configuration.

You could build your own pipeline to do this, but there are some available ones that you should be able to adapt to your own study design without too much difficulty. Some of these have matching downstream analysis tools that we will discuss later.

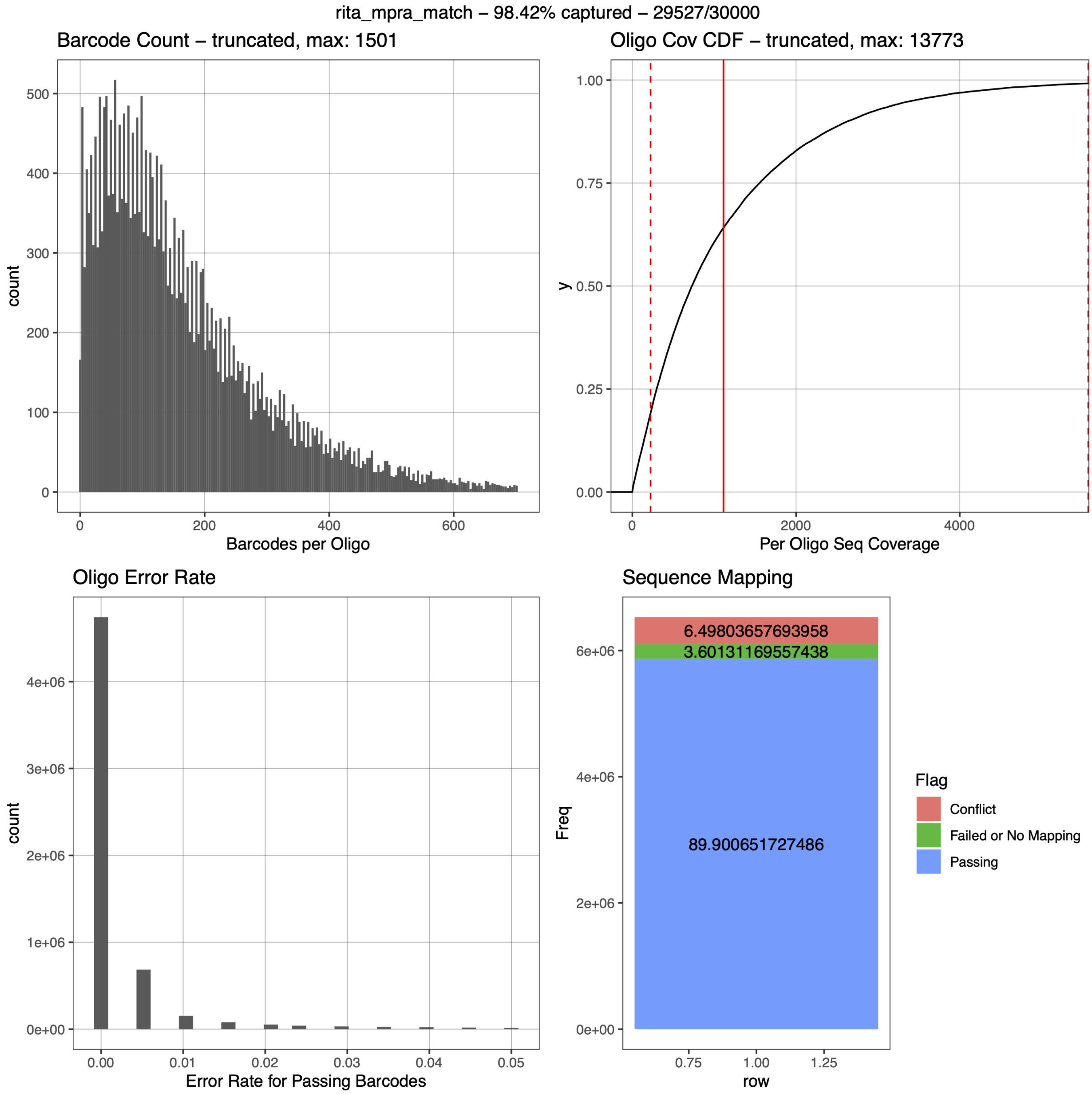
- MPRAFlow
- MPRASnakeflow
- MPRAmatch

Deep sequencing at this stage is essential to guarantee a good understanding of your library, so that there are no unassociated barcodes in the library when you transfect it, as these will have to be thrown out at data analysis stages. You can assess whether you have sequenced deep enough by plotting a saturation curve, in this style:



If additional reads result in no or very few novel barcodes, you have sequenced enough and can move on, provided you are satisfied with oligo representation, barcode uniqueness, and overall barcode-CRE association numbers.

This is an example of output from MPRAmatch, showcasing some basic QC for barcode matching. This library has a high number of barcodes!



Library transfection and data generation

As above, will not be covered today. As with library design, read read read, and ask people for help. If you are performing a lentiviral MPRA, Gordon et al, Nature Protocols, 2020 contains a comprehensive walkthrough every step from library design to data analysis (although it does not include more modern approaches such as the possibility of using a landing pad set up). Episomal MPRAs are simpler to carry out, but there is no equivalent detailed protocol available. Methods reportings in papers varies significantly in depth and detail, so it's hard to recommend any specific ones.

Raw data quantification (generating a counts table)

Will not be covered in detail. The pipelines above offer some degree of functionality towards doing this, but fundamentally you can use any mapper you like to link barcodes from either your DNA or RNA reads to the barcode-oligo association map you assembled in the `Associating barcodes to oligos` step, using that as a dictionary. Ideally, you can sequence *only* the barcode at this step, although depending on how flexible your sequencing provider is, it may actually be simpler to stick to a paired-end 150bp configuration or similar.

Barcode-level QC

This and the following section are the workhorses of MPRA data analysis, and where you should expect to spend most of your time. Some approaches and metrics are applicable at both the barcode and the oligo level; they are only mentioned in one of these sections but you will probably learn something interesting about your data if you carry them out twice.

If your design incorporates UMIs, you will want to incorporate that in the approach you take; that is currently beyond the scope of this repeat some of this at the UMI level, to ensure you are analysing unique sequences individually.

As with most barcoded assays, MPRA data is summarised as the log2 ratio of RNA-derived reads to DNA-derived pDNA reads, and can be analysed either at this level or treating RNA and DNA reads independently. The following two sections do a mix of both.

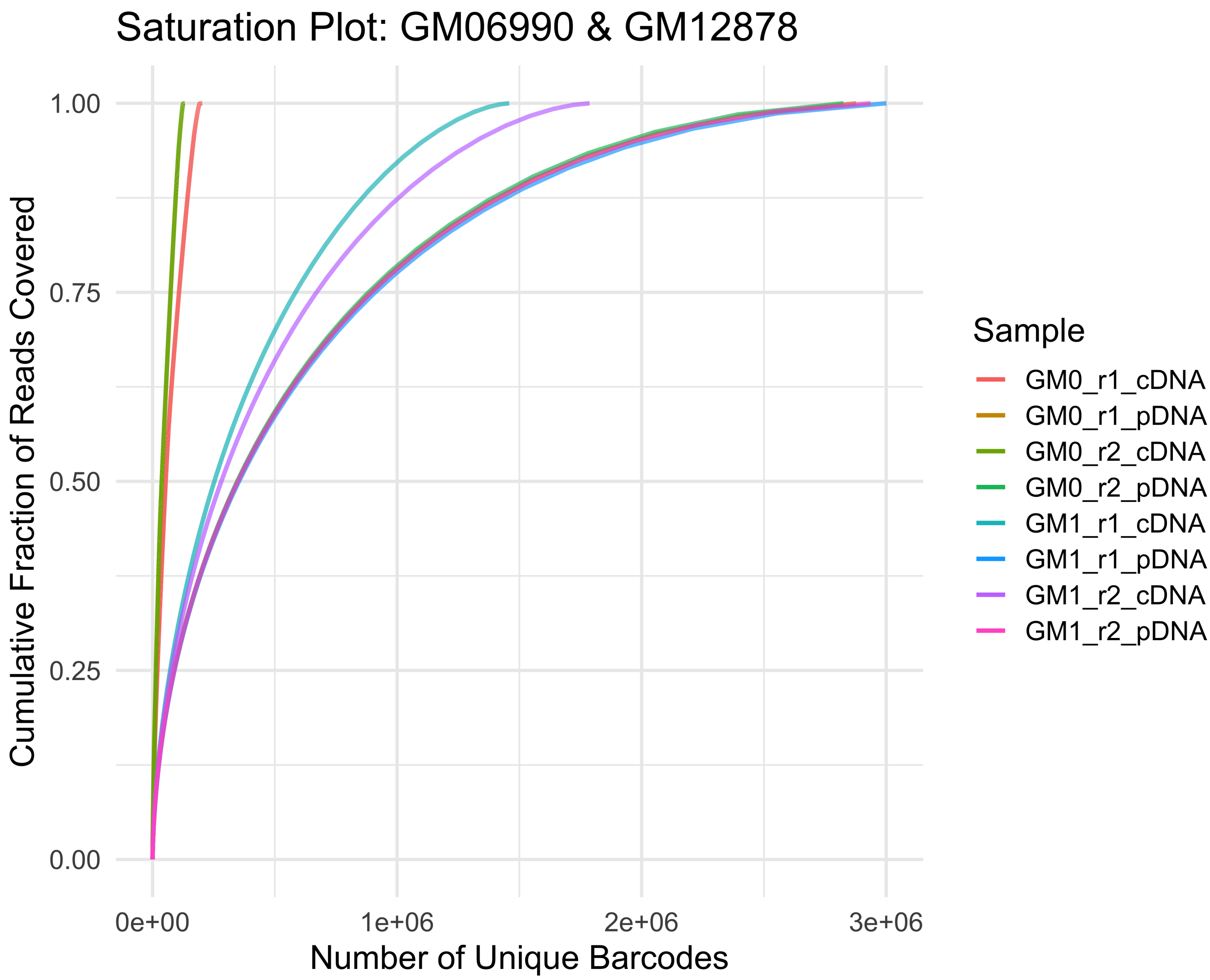
$$ \text{activity} = \log_2!\left(\frac{\text{RNA}}{\text{DNA}}\right) $$

Reads are likely to be further normalised by using a counts per million (CPM) transform. In addition, sometimes RNA and DNA will be referred to as cDNA and pDNA respectively, to denote that they are derived from reverse transcription of RNA and from the library plasmid pool.

Sequencing saturation and library complexity

Maintaining library representation throughout the experiment is crucial. Representation will affect the reliability of your replicates, with low complexity libraries yielding poor quality results.

As with the library-barcode association step, saturation curves are a straight-forward QC metric. The example below is from a mixtured of failed and successful transfection replicats—but how can you tell? (Note that there are multiple problems with this experiment!)



Kreevan et al, unpublished

PCA

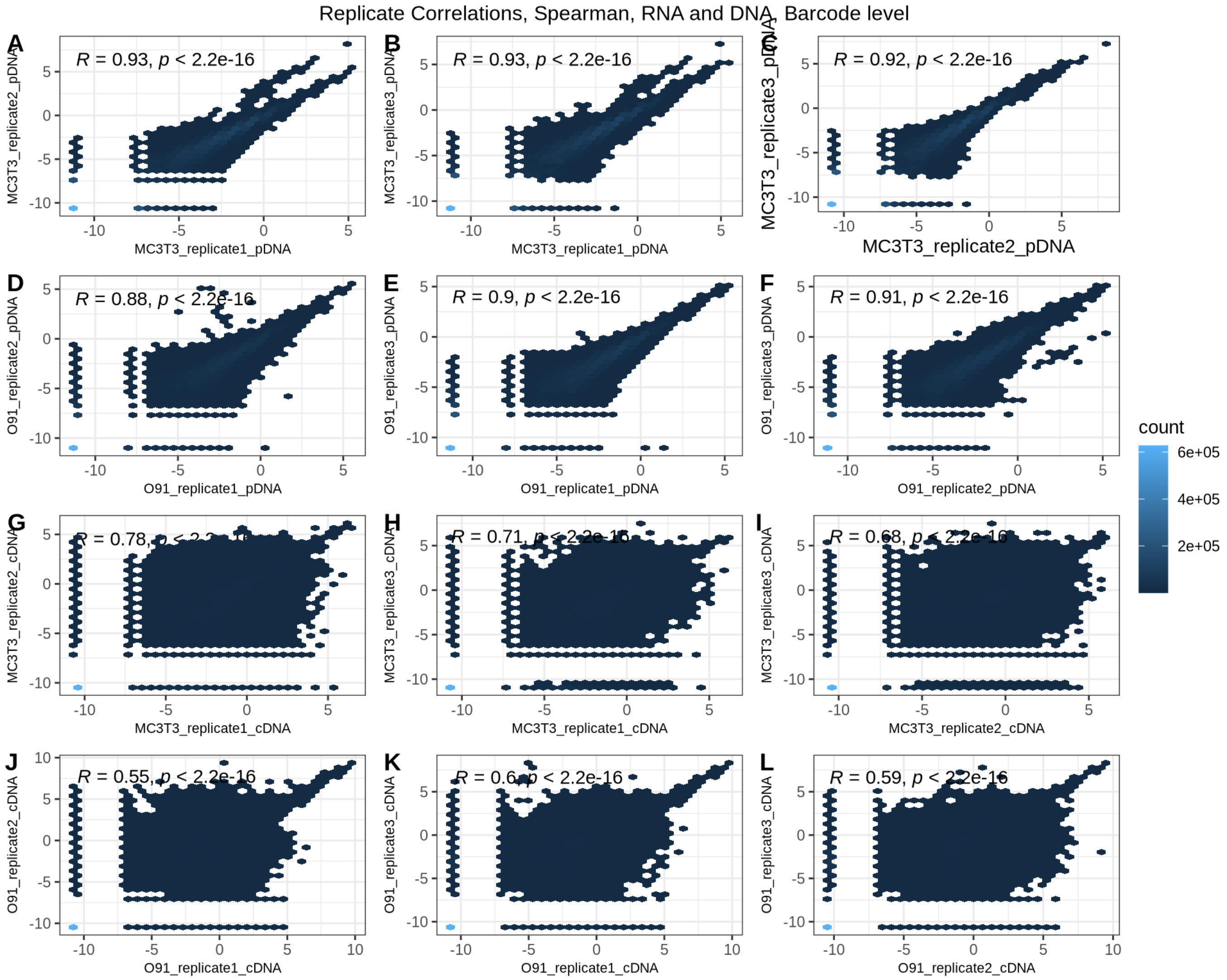
Principal component analysis of your dataset can be a quick and easy check of overall data quality, so it should be one of the first things you do. You can do this at the barcode or at the oligo level, after aggregation (see below), and you can analyse either DNA and RNA counts, or activity values. All of these will provide complementary information that will help you diagnose possible problems.

Checking for GC bias

Higher GC content is generally associated with higher transcriptional activity, all else being equal. Within an oligo, high GC barcodes may potentially be more active than low GC ones; at the aggregated oligo level high GC content oligos are more likely to be active than lower GC content oligos. Whether you implement a filter for this or simply sanity check to make sure there are no extreme outliers is entirely up to you, but it is worth looking into.

Correlations between and within replicates

This is essential, and can be assessed in many ways. It is common to examine the correlation between cDNA or pDNA levels across replicates, as in this figure:

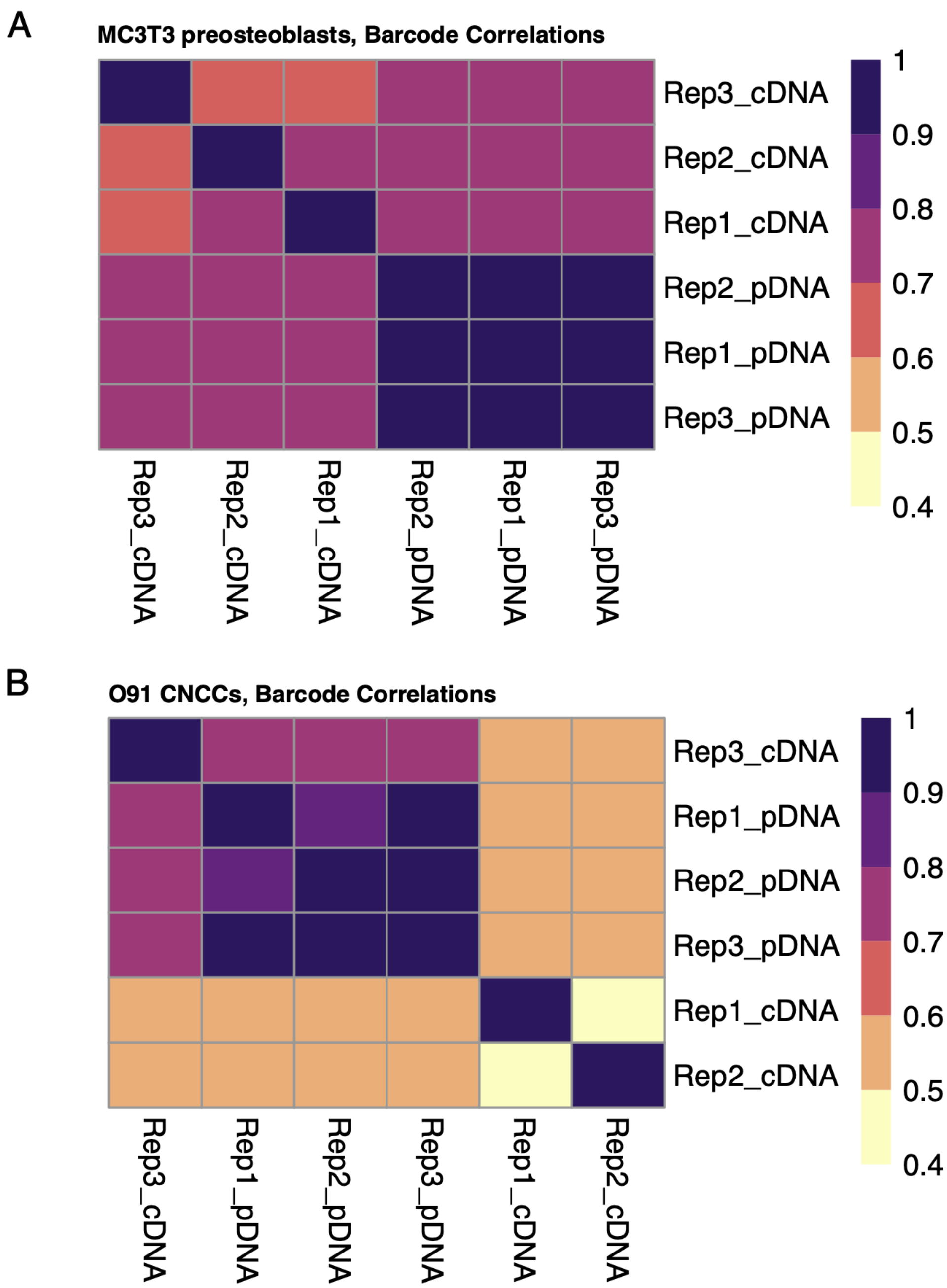


Shukla et al, bioRxiv, 2025

As shown below, correlation should be higher for DNA counts between replicates than RNA counts. RNA data will exhibit some correlation but can be a lot noisier, and this is driven both by very lowly active oligos that may make up the bulk of your pool, and by stochastic noise across replicates.

In addition, undersequenced experiments (see saturation curves above) will show poor correlations within a data type (DNA or RNA) across replicates, as well as between data types within replicates.

You can also examine the correlation between RNA and DNA within a single replicate. There are two clearly problematic libraries in the experiment below, and they are easier to identify this way than in the figure above—although you can spot them there too.



Shukla et al, bioRxiv, 2025

Other things to pay attention to include the dynamic range of your library, []

Oligo-level QC

Although you may perform your activity and differential activity testing at the barcode level, QC at the oligo level is still essential, since this is the level at which data becomes biologically interpretable. As such, familiarising yourself with the following QC steps is essential to making sure you can trust your results. It is worth noting that while some issues with experiments can generally be diagnosed in multiple different ways more subtle problem might only be apparent when you look at one specific metric, so even if these suggestions seem somewhat redundant, exhaustive QC always pays off. As above, many of these checks can be carried out at the barcode level *as well as* at the oligo level.

Data aggregation

To go from barcode-level data to oligo-level data you must aggregate your data in some way, even if it is only for QCing data. There are multiple ways of aggregating data, but the most commonly seen ones are

- Sum reads across barcodes
- Use the mean activity across barcodes
- Use the median activity across barcodes

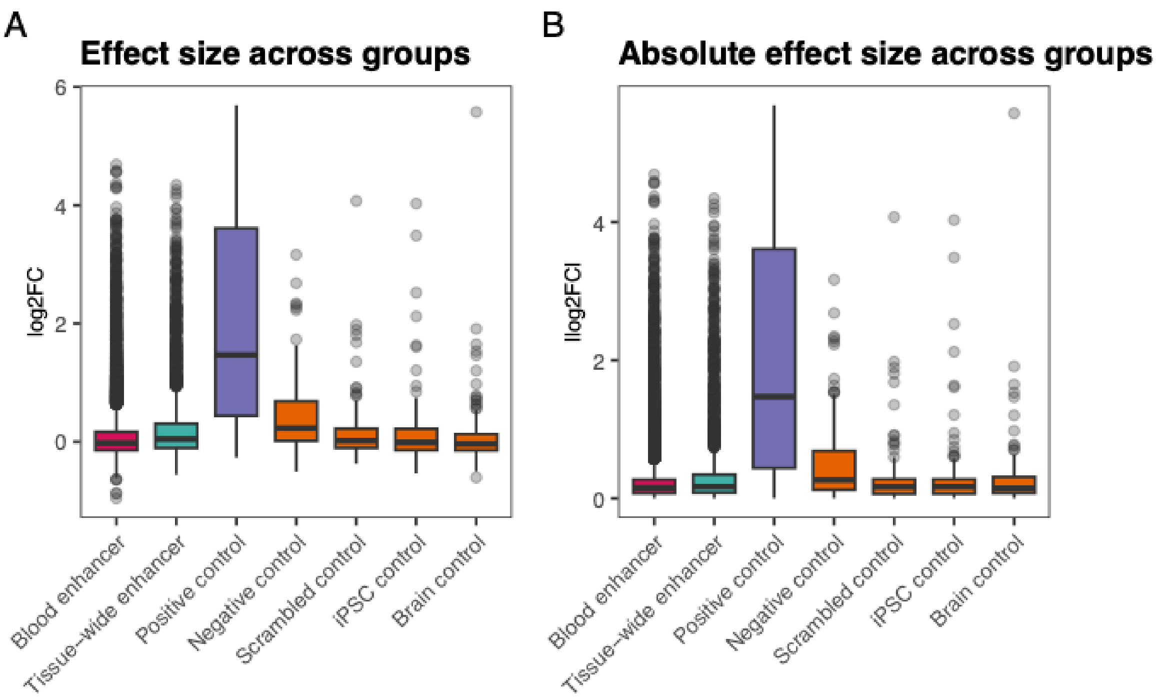
Mean and sum are the most popular, median is better protected against outlier barcodes. You may want to explore the barcode-level dataset below and examine the impact of different definitions on estimates of activity and what is retained for downstream analysis estimates.

You can also consider more sophisticated approaches, such as aggregating using a trimmed mean/after filtering outlier barcodes; this approach is better at stabilising variance and controlling for outlier than a simple mean.

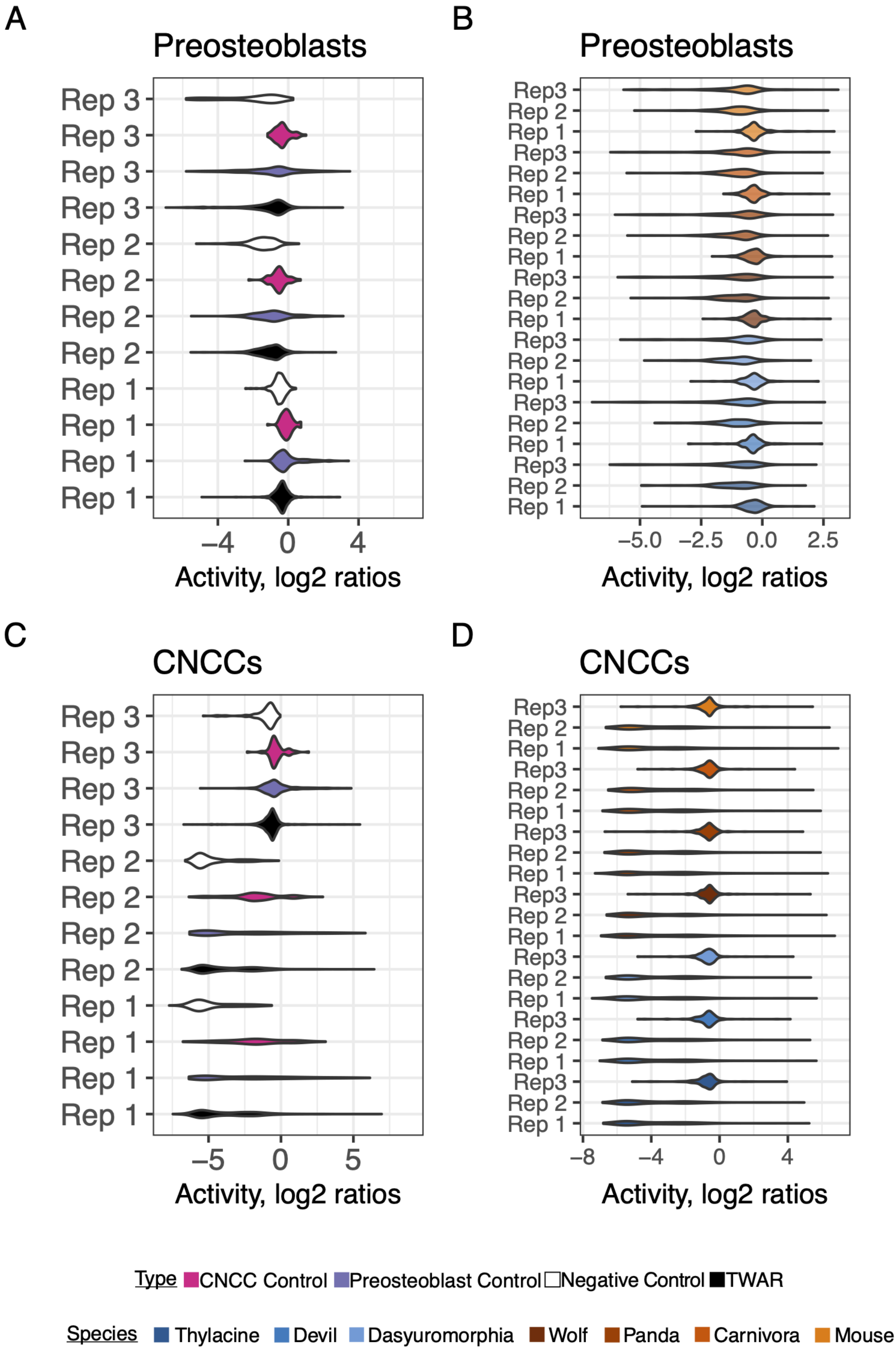
Examining control sequences

Careful checking of your control sequences is key at both barcode and oligo levels. You can use controls to calibrate thresholds for activity testing or simply as a sanity check that your experiment has a decent dynamic range and that you can proceed to formal activity testing using a more open-ended model.

Depending on the dsicriminatng ability of your control sequences and the nature of your experiment, the separation between positive and negative controls, or between controls and test sequences may vary a lot. For instance, consider the following two sets of figures, from two different MPRAs. What do you make of the trends in either experiment?



Kreevan et al, unpublished.

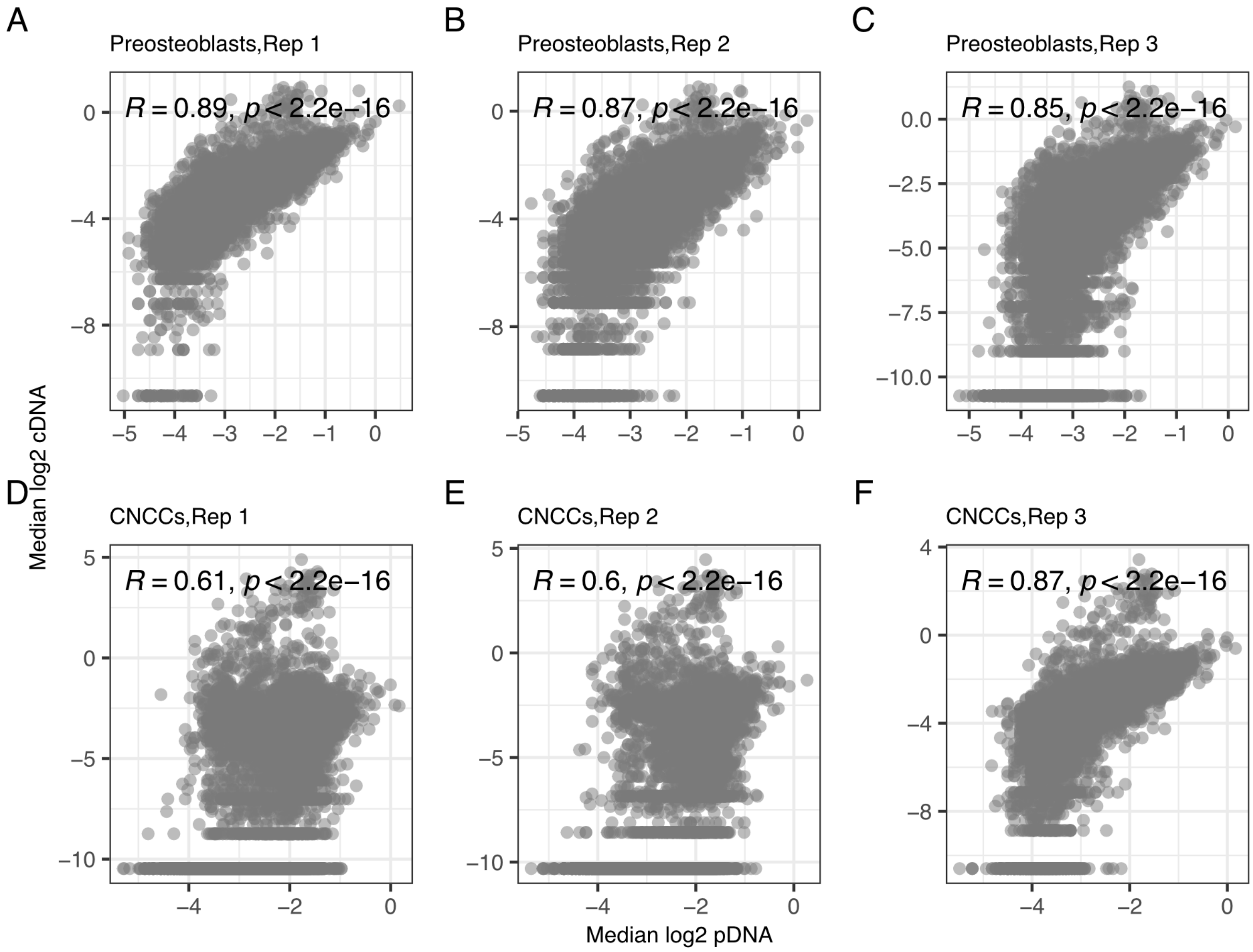


Shukla et al, bioRxiv, 2025.

Examining within replicate activity levels:

Visualising RNA vs DNA levels at the oligo level is essential to good validation.

In a successful experiment, the two data modalities will exhibit a good degree of correlation, but not perfect. Very high correlations (> 0.9 as a rule of thumb) suggest a failed experiment where RNA levels can be fully predicted by plasmid DNA levels. In the libraries below, you can appreciate cell-type specific effects on activity, which likely stem from reproducible cell-type specific effects on transfection efficiency. You can also appreciate a cluster of oligos in each plot with very high RNA levels relative to their DNA levels. Observing this is good reassurance that the experiment has been successful and it is worth analysing further.



Shukla et al, bioRxiv, 2025.

Statistical modelling: identifying activity and differential activity

As discussed in the lecture, there are many perils around defining a robust null model when performing MPRA data analyses. The nature of the non-coding genome makes this an inherently more challenging task than when analysing DMS or similar data sets drawn from protein-coding regions. This has implications for downstream analysis.

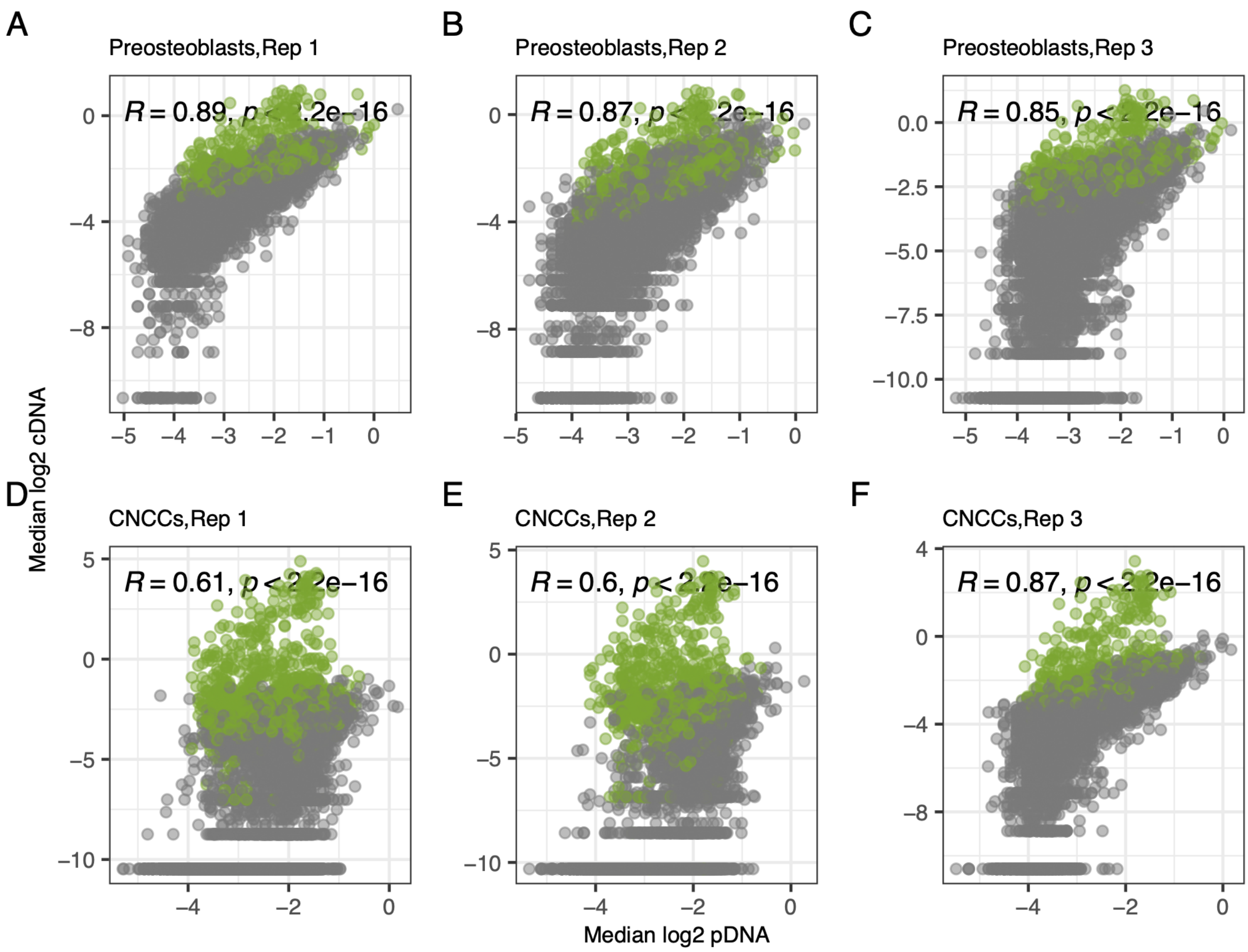
Depending on your study design, you may want to simply identify active elements from a set of tested elements, or you may want to identify differential activity within a given genomic element, across many possible genomic elements. These two require slightly different approaches.

Examining activity levels

Historically, the lack of robust statistical testing models for identifying active and differentially active elements in an MPRA meant that many experiments were analysed using a two-step design (see also note on filtering below). In the first step, a set of active sequences was defined using a metric, and in the second step, these sequences were tested one by one for differential activity.

For instance, [Uebbing et al, PNAS 2021] tests all oligos in their MPRA for activity by compapring activity levels across all barcodes to the dataset average using a one-tailed t test. Significant oligos (after multiple testing correction) were deemed and then tested for differential activity. Although this approach is coarse and likely too conservative, it provides a simple heuristic for filtering that is more independent of the behaviour of controls.

In the figure below, which is the same dataset as above, sequences deemed 'active' by this definition are coloured green. This probably represents a conservative subset of the 'true' set of active sequences.



Shukla et al, bioRxiv, 2025

These days, it is more common to skip activity testing altogether if you are looking for differentially active regions, or to explicitly incorporate activity testing into a linear model by including relevant terms in the design matrix. For an example of this, see Abell et al, Science, 2022, and its associated github repo, especially this notebook and the multiple specified design matrices. However, in this study, the authors explicitly test each oligo for a significant difference in RNA and DNA levels, rather than defining activity externally and using that as its data points:

$$
\begin{aligned}
i &= material\ (DNA, RNA) \\
j &= sample\ ID \\
k &= allele\ (ref, alt) \\
count_{ijk} &\sim \mu + \alpha_i + \beta_{i(j)} + \gamma_{i(k)} + \epsilon_{ijk}
\end{aligned}
$$

(Abell et al, Science, 2023)

> Here, α describes the aggregate allele-independent effect observed in RNA compared to DNA, while β and γ describe sample-specific and allele-specific effects respectively, nested within either RNA or DNA as denoted by the subscript parentheses. Using a linear contrast, we extracted the global RNA vs DNA average effect as “expression” effects, and significantly non-zero nested allelic terms indicate “allelic” effects. In this way, we obtained summary statistics that correspond well to the fold change and skew values reported in (8), respectively, but do so with an integrated model that does not require separate inference of allelic effects after expression effects (fig. S2).

Available tools:

Most tools for handling MPRA data are R-based, and generally built on an overdispersed negative binomial or Poisson model that is commonly used for analysing other NGS count data. These tools make a fundamental assumption is that most elements being tested are not going to be differentially active between states, and that your p-value distribution is not extremely skewed.

If you do not think your data will meet those assumptions, it's worth talking at lenght to a real statistician about the suitability of these approaches vs tools geared to another end. All of these methods can be run with unsuitable data, and it is ultimately on you to make sure you are aware of their assumptions and limitations - an at-length discussion of these is beyond the scope of this module! That said, the most established tools are *mpralm* and *MPRAnalyze*. *mpralm* is simpler and exponentially faster to run, but cannot easily accomodate complex study designs and only analyses data at the aggregate oligo level. It is built on the *edgeR* negative binomial model and makes uses of the empirical Bayes framework. *MPRAnalyze* will analyse data at the barcode level, but has very high computational requirements, and is very slow, so it is decreasing in popularity. A newer tool is *BCalm*, which is built upon *mpralm*, but can model the data at barcode levels.

Some options:

- For MPRAs and STARR-seq:
  - BCalm (Keukeleire et al, BMC Bioinformatics 2025)
  - MPRAnalyze (Ashuach et al, Genome Biology 2019)
  - mpralm (Myint et al, BMC Genomics 2019)
  - QuASAR-MPRA (Kalita et al, Bioinformatics 2018)
- For STARR-seq only (a mixture of preprocessing and analysis):
  - CRADLE (Kim et al, Genome Research 2021) (really better suited for preprocessing)
  - STARRPeaker (Lee et al, Genome Biology, 2020)

- More general count-based methods for data modelling (from RNA-seq, come with their own assumptions):
    - limma/voom (Ritchie et al, NAR, 2015; Law et al, Genome Biology, 2014)
    - edgeR (Chen et al, NAR, 2025)
    - DESeq2 (Love et al, Genome Biology, 2014)
- More artisanal (mostly deprecated):
    - One or two-step ANOVA designs (as in Uebbing et al, PNAS, 2021, Shukla et al, bioRxiv, 2025)

The limma user guide also contains a solid tutorial on putting together linear model design matrices, which you should consult any time you're trying to analyse count-based data. Another valuable resource is Modern Statistics for Modern Biology, by Susan Holmes and Wolfgang Huber, Cambridge University Press (2019), which can be fully accessed for free and also contains a general overview on analysing NGS count datasets.

A note on filtering elements:

To limit the number of statistical tests you carry out and therefore decrease your testing burden, you may want to discard oligos or barcodes (depending on your analysis approach) where you know, *a priori*✅ reads or cpm). Another approach is to actively test each oligo against the positive controls under the assumption that those represent the true distribution of active values—but that requires a robust positive control distribution in the first place and is likely to be too stringent—or only test oligos in the top [x]% of the activity distribution.

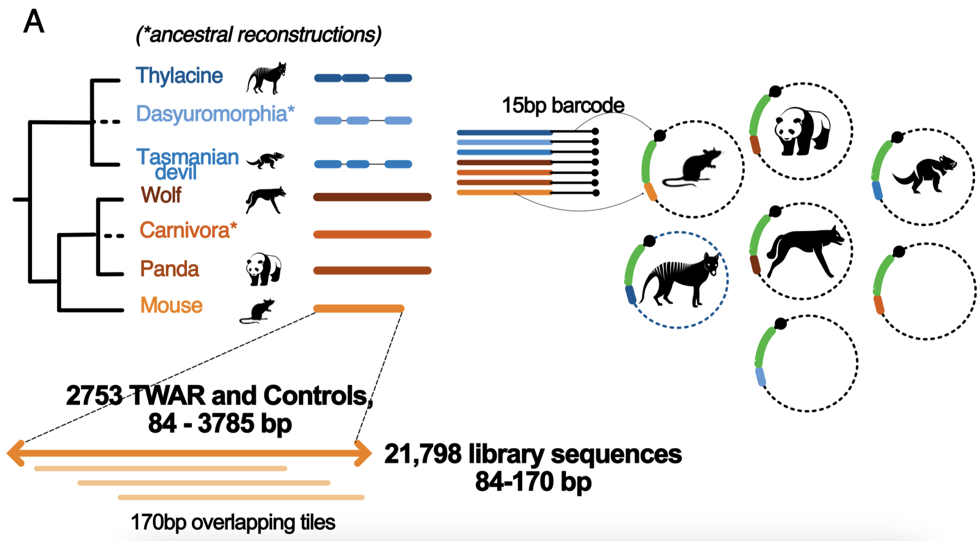
Defining inactive oligos and what you have power to detect is contingent on the expected effect distribution of effect sizes in your datasets. Filtering that is too generous will result in wasted tests, whereas filtering that is too stringent (eg, matches the distribution of controls in the figure above) will lead to too few elements being retained and is likely to cause any count-based modelling tool to falter, and give a poor idea of library complexity. If you have enough barcodes and replicates you may wish to skip filtering altogether and proceed straight to activity (and/or differential activity) testing.

What now?

Explore one of the following datasets, taken from Comerford et al, bioRxiv, 2025. They are taken from an episomal MPRA study testing for regulatory activity across ~30,000 introgressed SNPs from Neanderthal and Denisovans segregating in individuals of Papuan-like genetic ancestry at allele frequencies > 0.15 and located within annotated immune enhancers. The dataset is available either as:

- MPRAfrag1.txt: Barcode-level data across 5 replicates
- MPRAnorm1.txt: Oligo-level data across 5 replicates

Alternatively, if you want to explore a failed experiment (and attempt to diagnose what went wrong), you can look at the file below. This is the same library and set up that is described in Shukla et al, bioRxiv, 2025, but as a failed experiment the data was not included in the final publication. The study design is complex, and not a 'traditional' MPRA. Instead, it tests a library of 21,987 unique elements consisting of orthologous convergently evolutionarily accelerated sequences from 7 taxa (thylacine, grey wolf, Tasmanian devil, giant panda, Dasyuromorphia, Carnivora and mouse) and positive and negative controls in two cell lines (MC3T3-E1 preosteoblasts and O91 cranial neural crest cells) as detailed in the schematic below.



(Shukla et al, bioRxiv, 2025)

- shukla_et_al_bad_library.count: Barcode-level data across 3 replicates across 2 cell lines

No matter the dataset, you will have to download it from the google drive link above and read it into your choice of Jupyter or R notebook. This document provides only rough code examples for analysis, so you will have to do some adjusting yourself depending on what you choose to do! Things you might consider doing include:

- Explore the datasets and use whatever descriptive statistics you want to come to grips with them. They may not be in standard formats, so you might have to use your existing data science skills to clean them up/reshape them into a more straightforward format.
- Perform PCA on any of the datasets above to quickly diagnose any issues
- Apply some of the barcode or oligo level QC metrics discussed above
- Explore the consequences of different aggregation approaches
- Use the negative and positive controls to filter test regions. Does this seem like a good approach? Why or why not?
- Use a simple linear model or ANOVA to attempt to identify differentially active elements in any of the assays

Alternatively, you may also explore the complete analyses underpinning these papers by examining the associated workflowR notebooks:

- Comerford et al Denisovan introgression MPRA
- Shukla et al Thylacine-wolf convergence MPRA

Note that these are not authoritative nor perfect, and indeed they do not necessarily reflect ideal experimental designs, but rather works in progress. Both lead analysts have made decisions driven by the nature of their own data and their experimental conditions that may or may not be relevant to you and your study design. However, exploring their notebooks will help you familiarise yourself with the kinds of analyses done at every step of the process.